

Hard Water - Study Notes

1. Definition:

Hard water is water that contains a high concentration of dissolved minerals, mainly calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions.

2. Types of Hardness:

Temporary Hardness Caused by bicarbonates of calcium and magnesium. It can be removed by boiling.

Permanent Hardness Caused by sulfates and chlorides of calcium and magnesium. It cannot be removed by boiling.

3. Causes of Hard Water:

Water passing through limestone and chalk deposits.

Dissolution of calcium and magnesium salts from rocks.

4. Disadvantages of Hard Water:

Reduces soap lather formation.

Forms scale in boilers and pipes.

Increases energy consumption in heating systems.

Can damage appliances.

5. Advantages of Hard Water:

Provides essential minerals like calcium and magnesium.

May contribute to better taste in drinking water.

6. Methods of Softening Hard Water:

Boiling (for temporary hardness).

Adding washing soda (Na_2CO_3).

Ion-exchange (zeolite/permutit process).

Reverse osmosis method.

7. Measurement of Hardness:

Measured in ppm (parts per million) or mg/L as CaCO_3 equivalent.

Determined using EDTA titration method in laboratories.